

ZPJA PROJECT REPORT

Building sentence encoders from large language models

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Ondřej Bahounek (xbahou00@vutbr.cz)

Jiří Tomášek (228749@vutbr.cz)

1 Introduction

Sentence embeddings aim to represent variable-length text sequences as fixed-dimensional vectors, such that semantic similarity between sentences is preserved in the embedding space. These representations are a key component in tasks such as semantic textual similarity (STS), clustering, retrieval, and information matching.

Several established methods, including SBERT and SimCSE, demonstrate that fine-tuning transformer encoders with contrastive or classification objectives can substantially improve the quality of sentence representations. Building on these ideas, this project systematically studies how different **pooling strategies** and **training objectives** affect the quality of sentence embeddings derived from BERT-based encoders.

Our primary goal is to evaluate whether more complex pooling mechanisms, such as attention-based pooling, provide measurable benefits over simpler baselines like mean pooling.

2 Model Architecture

We use `bert-base-uncased` as the backbone encoder. Given an input sentence, BERT produces contextualized token embeddings:

$$H = [h_1, h_2, \dots, h_n], \quad h_i \in \mathbb{R}^d$$

A pooling layer is applied on top of these token embeddings to obtain a fixed-size sentence embedding $s \in \mathbb{R}^d$.

2.1 Pooling Strategies

We evaluate the following pooling methods:

- **Mean Pooling:** Average of all token embeddings.
- **Max Pooling:** Element-wise maximum over token embeddings.
- **Last-Layer Attention Pooling:** Learnable multi-head attention over the final hidden layer.
- **Multi-Layer Attention Pooling:** Attention pooling applied across multiple hidden layers.

2.2 Architecture Overview

Figure 1 illustrates the overall architecture consisting of a BERT encoder followed by a pooling layer.

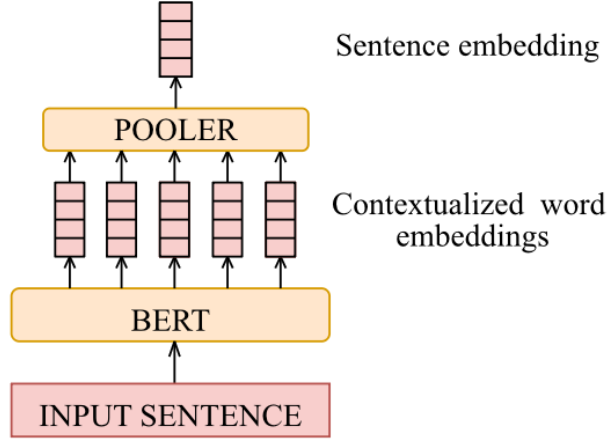


Figure 1: BERT encoder followed by a pooling layer producing a sentence embedding.

2.3 Attention Pooling

In contrast to static aggregation methods such as mean or max pooling, **attention pooling** allows the model to assign different importance weights to individual tokens, potentially emphasizing semantically informative words while down-weighting less relevant ones.

We implement a **multi-head attention pooling** (see fig. 2) mechanism operating on the token-level embeddings produced by the transformer encoder. For each head, a fixed number of **trainable query vectors** is learned, which attend over the token embeddings to produce multiple pooled representations. The resulting vectors from all heads are concatenated and passed through a small **MLP** to reduce dimensionality and produce the final sentence embedding.

In the **Multi-Layer Attention Pooling** variant, token embeddings from multiple transformer layers are first combined using a learned weighted average. This layer aggregation is motivated by prior findings that different transformer layers capture complementary linguistic information, ranging from syntactic to semantic features. The aggregated embeddings are then provided as input to the attention pooling module.

3 Training Objectives

We experiment with three loss functions commonly used for sentence embedding learning.

3.1 Triplet Loss

Triplet loss enforces a margin between positive and negative pairs:

$$\mathcal{L}_{triplet} = \max(\|s_a - s_p\| - \|s_a - s_n\| + \epsilon, 0)$$

3.2 Multiple Negatives Ranking (MNR) Loss

MNR loss uses all other samples in the batch as negatives:

$$\mathcal{L}_{MNR} = -\log \frac{e^{\text{sim}(s_a, s_p)/\tau}}{\sum_{j \in \mathcal{B}} e^{\text{sim}(s_a, s_j)/\tau}}$$

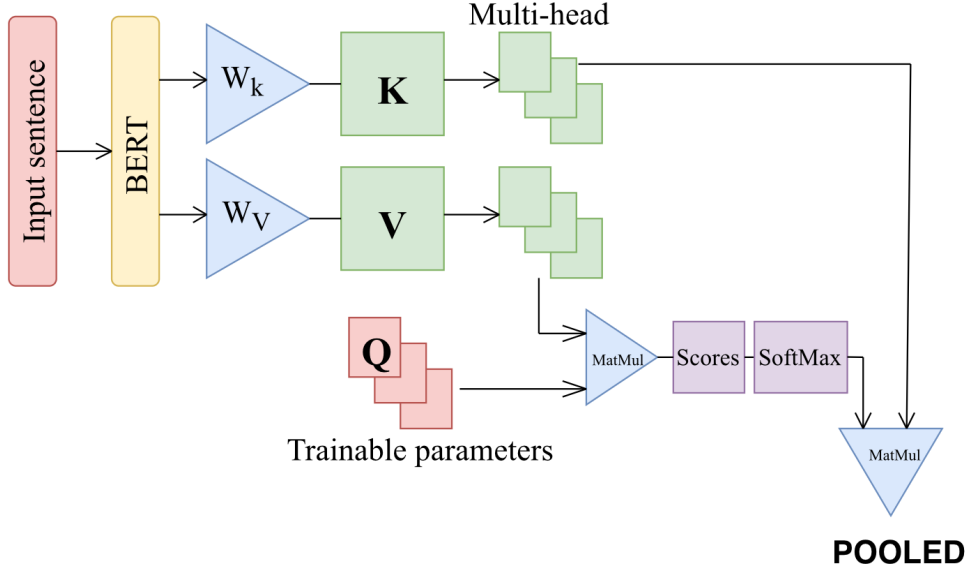


Figure 2: Structure of the attention-based pooling mechanism.

This objective benefits from larger batch sizes, as they increase the number of negative examples.

3.3 SBERT Classification Loss

Following the original SBERT approach, sentence pairs are encoded independently and combined using a matching heuristic before classification. Given sentence embeddings u and v , the classifier input is formed as

$$h = [u; v; |u - v|]$$

and passed through a linear layer followed by a softmax.

The model is trained using cross-entropy loss over Natural Language Inference (NLI) labels:

$$\mathcal{L}_{SBERT} = \text{CE}(\text{softmax}(Wh + b), y)$$

where $y \in \{\text{Entailment, Neutral, Contradiction}\}$ is the ground-truth NLI label.

After training, the classifier head is discarded and the sentence encoder is used to produce fixed-size sentence embeddings.

4 Experimental Setup

- **Training Data:** SNLI and MNLI datasets (over 500k sentence pairs).
- **Encoder:** bert-base-uncased.
- **Optimization:** AdamW optimizer with learning rate $2e-5$.
- **Evaluation:** STS Benchmark using Spearman rank correlation.

5 Results

We report performance using Spearman’s rank correlation coefficient $\rho * 100$ on the STS Benchmark.

5.1 Impact of Loss Function

We compared different training objectives while keeping the pooling fixed to mean. Our results show that MNR loss outperforms all other losses.

Table 1: Effect of training objective using mean pooling.

Loss Function	STS ρ
SBERT Classification	75.6
Triplet Loss	77.4
MNR Loss (Batch Size = 32)	80.9
MNR Loss (Batch Size = 64)	82.0

5.2 Impact of Pooling Strategy

We compared different pooling strategies while keeping the training objective and batch size fixed. In our experiments, mean pooling achieved the best performance.

Table 2: Effect of pooling strategy using MNR loss with batch size 32.

Pooling Method	STS ρ
Mean Pooling	80.9
Max Pooling	78.8
Last-Layer Attention Pooling	79.6
Multi-Layer Attention Pooling	80.2

5.3 Attention Pooling Layer Results

We investigate the performance of attention pooling on sentence embeddings, varying the number of attention heads and trainable query vectors per head.

5.3.1 Last-Layer Attention Pooling

Table 3 shows results for attention pooling applied to the last layer of the transformer. Different configurations of heads and queries per head were tested.

5.3.2 Multi-Layer Attention Pooling

Table 4 reports results for multi-layer attention pooling, where embeddings from several transformer layers are first aggregated (weighted average) before applying attention pooling. This allows the model to leverage features from multiple depths of the transformer.

Table 3: Last-layer attention pooling with varying heads and queries per head.

Heads	Queries per Head	STS ρ
8	8	79.5
8	16	79.6
8	32	79.1
8	64	78.7
16	8	78.5
16	16	78.6

Table 4: Multi-layer attention pooling with varying heads, queries, and number of aggregated transformer layers.

Heads	Queries per Head	STS ρ
16	8	78.8
8	4	79.9
8	1	80.2
8	1	79.3

From these experiments, we observe that lightweight configurations (fewer queries, moderate heads) often achieve the best trade-off between model complexity and performance, particularly for multi-layer attention pooling.

5.4 Baselines and Reference Models

For comparison, we include standard pre-trained sentence embedding models and a non-fine-tuned BERT baseline. These provide a point of reference for the performance of our trained models.

Table 5: Baseline and reference model performance on STS Benchmark.

Model	STS ρ
BERT-base (mean pooling, no fine-tuning)	46.3
SBERT-base (Reimers et al., 2019)	77.3
SimCSE-BERT-base (Gao et al., 2021)	84.2

6 Conclusion

In this project, we evaluated several pooling strategies and loss objectives for learning sentence embeddings. Our experiments indicate that while attention-based pooling offers additional modeling flexibility, simple mean pooling remains a strong and competitive baseline. Among the evaluated objectives, Multiple Negatives Ranking loss achieved the best overall performance, particularly when trained with larger batch sizes.